

VILJANDI, Estonia

WMO: 262330

Lat: 58.3778N

Lon: 25.6003E

Elev: 88

StdP: 100.28

Time Zone: 2.00 (EUE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.6	-17.7	-23.9	0.4	-21.4	-20.1	0.6	-17.2	7.2	1.8	6.5	0.9	1.3	60	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	27.9	19.6	26.1	18.3	24.3	17.5	20.7	25.7	19.5	24.2	18.5	22.9	2.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.8	22.7	17.8	12.9	21.8	16.7	12.0	20.6	59.8	25.7	55.9	24.0	52.5	22.9	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.1	5.3	4.7	DB	-24.0	30.3	4.8	2.0	-27.5	31.7	-30.3	32.8	-33.0	33.9	-36.5	35.3	(4)
(5)			WB	-24.0	22.2	4.5	1.1	-27.3	23.0	-29.9	23.6	-32.4	24.2	-35.7	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-5.1	-4.9	-0.8	5.4	11.7	15.2	18.1	16.8	12.0	5.8	1.6	-1.7	(6)
(7)		DBStd	9.19	6.51	6.19	4.65	3.79	4.17	3.29	3.06	2.91	3.14	4.09	4.05	5.38	(7)
(8)		HDD10.0	2162	468	416	336	146	28	2	0	0	17	137	251	361	(8)
(9)		HDD18.3	4479	726	649	594	389	210	105	42	66	190	387	501	619	(9)
(10)		CDD10.0	792	0	0	0	7	81	157	252	211	77	8	0	0	(10)
(11)		CDD18.3	67	0	0	0	0	4	10	36	18	0	0	0	0	(11)
(12)		CDH23.3	574	0	0	0	0	45	102	287	139	2	0	0	0	(12)
(13)		CDH26.7	106	0	0	0	0	4	15	61	25	0	0	0	0	(13)
(14)	Wind	WSAvg	2.2	2.5	2.4	2.5	2.3	2.0	2.1	1.8	1.8	1.8	2.2	2.5	2.8	(14)
(15)	Precipitation	PrecAvg	562	39	28	27	28	29	36	47	61	66	64	69	53	(15)
(16)		PrecMax	770	76	65	63	70	86	95	117	184	163	154	132	111	(16)
(17)		PrecMin	413	6	1	2	1	4	0	14	0	20	13	18	6	(17)
(18)		PrecStd	83	19	17	13	17	18	22	27	39	34	34	26	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	5.7	13.5	20.9	27.0	28.6	30.2	29.3	22.8	15.7	9.8	8.8	(19)
(20)			MCWB	5.4	4.0	6.7	10.7	16.7	19.1	20.8	20.0	16.7	12.6	8.1	7.8	(20)
(21)		2%	DB	4.6	4.2	9.2	17.7	24.4	26.2	28.6	26.6	20.3	13.8	8.7	7.0	(21)
(22)			MCWB	4.0	3.2	5.4	9.4	16.1	17.4	19.9	18.9	15.3	12.0	7.8	6.1	(22)
(23)		5%	DB	3.2	2.9	6.6	15.3	22.1	24.0	26.6	24.6	18.7	12.6	7.8	5.6	(23)
(24)			MCWB	2.5	2.2	3.6	8.3	14.4	16.9	19.2	17.9	14.6	11.3	7.0	4.8	(24)
(25)		10%	DB	2.1	1.9	4.7	12.7	19.8	21.8	24.7	22.6	17.2	11.5	6.8	4.3	(25)
(26)			MCWB	1.6	1.3	2.7	7.0	13.1	15.3	18.4	17.0	13.9	10.4	6.2	3.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	4.6	7.5	11.7	19.7	20.5	22.4	21.7	17.4	13.5	8.8	7.9	(27)
(28)			MCDWB	6.1	5.2	12.2	17.9	24.6	25.9	27.9	26.3	20.7	14.9	9.3	8.4	(28)
(29)		2%	WB	4.1	3.4	5.5	10.1	17.3	18.9	21.1	19.9	16.2	12.3	8.0	6.3	(29)
(30)			MCDWB	4.4	4.1	8.4	16.5	22.3	24.2	26.3	24.6	19.1	13.3	8.5	6.9	(30)
(31)		5%	WB	2.6	2.3	4.1	8.9	15.6	17.5	20.1	18.8	15.2	11.6	7.2	4.8	(31)
(32)			MCDWB	3.1	2.8	6.0	14.0	20.3	22.3	24.7	22.9	17.9	12.5	7.7	5.4	(32)
(33)		10%	WB	1.6	1.4	3.1	7.6	13.9	16.1	19.0	17.8	14.3	10.5	6.2	3.7	(33)
(34)			MCDWB	2.0	1.9	4.6	11.7	18.6	20.8	23.8	21.7	16.5	11.4	6.7	4.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	5.1	6.9	9.4	10.5	9.4	9.2	9.0	7.6	5.7	3.5	3.6	(35)
(36)			MCDBR	3.5	4.2	9.2	14.1	13.7	12.8	12.0	12.3	10.1	6.5	4.0	3.7	(36)
(37)			MCWBR	3.4	3.4	5.8	7.4	6.4	5.6	4.8	5.4	5.7	4.8	3.7	3.5	(37)
(38)		5% WB	MCDWR	3.6	4.0	8.3	12.2	11.9	11.3	10.8	10.7	8.6	5.7	3.9	3.5	(38)
(39)			MCWBR	3.5	3.4	5.7	7.0	6.2	5.8	4.8	5.3	5.2	4.5	3.7	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.305	0.334	0.370	0.364	0.364	0.389	0.386	0.354	0.334	0.314	0.269	(40)	
(41)		taud	2.129	2.151	2.215	2.312	2.387	2.414	2.362	2.369	2.444	2.403	2.222	2.117	(41)	
(42)		Ebn at Noon	541	707	801	836	868	873	838	812	784	685	491	420	(42)	
(43)		Edh at Noon	50	80	100	108	107	106	110	102	81	64	47	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.34	1.00	2.43	3.96	5.37	5.54	5.25	4.16	2.64	1.22	0.38	0.21	(44)	
(45)		RadStd	0.07	0.19	0.31	0.44	0.55	0.41	0.50	0.42	0.30	0.17	0.06	0.04	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-138	N/A	N/A	N/A	

Nomenclature: See separate page